

52%



A - 15
B - 11

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THE COPPERBELT UNIVERSITY

SCHOOL OF MEDICINE

End of term test: March 2019
MBS 240

Term 1- Laboratory Sciences

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Program: MBChB

Time: 1 hr.

Instructions:

Section A: Multiple choice Questions (20 marks).

Choose one correct answer and circle the letter in front of it.

There are 20 questions

Answer all questions

Section B: Short answers (30 marks).

Write short notes on 3 out of the 5 questions.

10 Marks are allocated to each question

Write your Student ID number on top of each page of the answer book.

Section A. Multiple choice questions.

1. An ordinary light microscope can be converted into a fluorescent microscope by using:

- a) Sunlight
- b) Normal light bulb light
- ✓ c) Ultra violet light
- d) Electron beam
- ⓔ None of the above

2. What term is given to the shortest distance between two points on a specimen that can be distinguished as separate entities under the microscope.

- a) Contrast
- b) Magnification
- c) Shrinking
- ✓ d) Resolution
- e) None of the above

3. What is the limit of resolution of unaided eye?

- a) 0.1 μm
- b) 0.1 cm
- ✓ c) 0.1 mm
- d) 0.5 μm
- ⓔ None of the above

4. In Electron microscope the electron beam is focused by:

- a) Objective lens
- b) Condenser
- c) Ocular lens
- ✓ d) Electromagnetic coil
- e) None of the above

5. Which one of the following is TRUE about a scanning Electron microscope?

- a) It was the first developed electron microscope
- ✓ b) It gives three dimensional objects in high resolution
- c) It is suitable for objects with moisture
- d) Object is in form of a thin slice of a sample
- e) None of the above

6. In which symbiotic relationship is there physiologic dependency of an organism on the host without the host benefiting from the relationship?

- a) Phoresis
- b) Mutualism
- ☒ c) Parasitism
- d) Commensalism
- e) None of the above

7. The largest overlap between symbiotic relationship is between:

- ☒ a) Parasitism and mutualism
- b) Phoresis/commensalism and Parasitism
- c) Mutualism and Phoresis/commensalism
- d) All of the above
- e) None of the above

8. Serum is:

- ☒ a) Same as plasma
- b) Is plasma without electrolytes
- ☒ c) Is plasma without fibrinogen
- d) Is plasma without urea
- e) None of the above

9. The earliest recognizable precursors of platelets are:

- ☒ a) Pronormoblasts R_{EC}
- ☒ b) Megakaryocytes
- c) Myeloblasts W_{EC}
- d) Basophils
- e) None of the above

10. What is the normal ~~proportion of solid part of~~ blood?

- ☒ a) 100%
- b) 30%
- c) 50%
- d) 60%
- ☒ e) 45%

11. Which degree of biochemical pathology is detectable in the resting state of a patient?

- a) Minor damage
- ☒ b) Major damage
- c) Hidden damage
- d) All of the above
- e) None of the above

12. The main body compartment in compartment analysis is:

- ☒ a) Intracellular fluid
- b) Pleural fluid
- c) Interstitial fluid
- ☒ d) Plasma
- e) None of the above

13. Secondary antibody responses are better because:

- ☒ a) They provide defence against unrelated antigens
- b) The antibody can be made by both T and B cells
- c) Complement-fixing antibodies are made
- d) They do not require T-cell help
- ☒ e) They are stronger and faster

14. Which cell type produces antibodies?

- a) Macrophages
- ☒ b) T-lymphocytes
- c) Natural Killer cells
- ☒ d) Plasma cells
- e) Eosinophils

15. The secondary, but not the primary, immune response is based on:

- ☒ a) Memory
- b) The toxin effect of multimerism
- ☒ c) Complement activation
- d) Mast cell degranulation
- e) Clonal selection

16. Which of the following statements does not apply to IgG?

- a) Appears early in the primary immune response.
- ☒ b) Neutralizes bacterial toxins.
- c) Can fix complement.
- d) Crosses the human placenta.
- ☐ e) Opsonizes bacteria.

17. Which of the following is true about a facultative anaerobe?

- ☒ a) Strictly require O_2 for their growth
- b) Require increased concentration of CO_2
- c) Cannot use oxygen for growth but grows in its presence
- ☐ d) Capable of growing either in presence or absence of O_2
- e) Require increased concentration of CO_2

18. During acid fast staining, Carbol fuchsin acts as

- ☒ a) Primary stain
- b) Mordant
- c) Counterstain
- d) Secondary stain
- e) Decolourizer

19. Which of the following antibodies is useful in diagnosis of acute viral infection?

- ☒ a) Ig A
- b) Ig E
- ☐ c) Ig M *1st*
- d) Ig G
- e) None of the above

20. The following are used for viral classification except

- a) Morphology
- b) Antigenic properties
- c) Genome type
- d) Replication method
- ☒ e) Cultural characteristics

Section B. Short answers.

1. Define the term "Neglected Tropical Diseases". What are the characteristics of these diseases? Give 5 examples of these diseases.
2. What is the relevance of requesting laboratory tests or investigations on a patient? *
3. Define the term "Reference range". What factors affect the reference range?
4. a) Describe the role of the skin as part of the innate immune system (5 marks)
b) Differentiate between Humoral and Cell mediated adaptive immune responses (5 marks)
5. List five types of media used in bacteriology and briefly describe each. (10marks)

Q4) 9) Skin plays various role in innate immunity such as
which are: Skin acts a barrier
Anatomical
- It

a) Skin plays a variety of roles. Some which include:
Skin acts as a barrier.

Anatomical

- Skin prevents the entry of household chemicals.
- Production of mucus in trachea traps microbes or small organism which are removed out.
- Cilia in the alimentary canal also traps microbes and propels them out.

Physiological

- pH
- Acid balance
- Metabolism

b) Humoral

Microbes are targeted captured and transported to immune cells that further kill and degrade them.

Cell mediated

There is activation of cells that ingest and degrade the microbes, these cells include macrophages, neutrophils and

1) These are diseases found in poverty stricken countries

- diseases that kill
- diseases that degenerate or cause disabilities
- diseases that cause blindness

Examples

Marasmus, Kwashiorkor

① Scurvy

② Kwashiorkor

③

1) These are diseases found in poverty stricken countries.
Some of the characteristics of these diseases are:

- ① diseases that kill
- ② diseases that can lead to blindness
- ③ diseases that can cause disabilities
- ④ diseases that can cause prolonged illness

And some of the examples of these diseases include:
Marasmus, Scurvy, Kwashiorkor

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② Laboratory tests are tests conducted on the patient to know what's wrong with the patient. It's relevant to that it helps ^{physicians} answer clinical questions like what's wrong with the patient, what's the cause of the problem, what are the effects of the problem over time and to find the best way in which to help the patient. To answer the question what's wrong with the patient history has to be taken and also laboratory tests have to be conducted. If the cause is known the question what's the cause of disease can be answered by taking differential diagnosis, and if the cause is known physicians can ^{now stage} a way of which they can help the patient.

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